GRADE 10 **MONDAY 2 FEBRUARY 2026**

MATHEMATICS CYCLE TEST

TIME: ± 1 hour **TOTAL: ±50 MARKS**

WHAT YOU NEED TO KNOW

ALGEBRA

Booklet up to and including page 9, as well as MAS exercises referenced in the booklet on pages 1-9.

1. Products
2. Factorising up to and including DOTS

GEOMETRY:

Booklet up to and including page 14, as well as:
MAS page 246 - 249 Ex 1 (a) – (e),
MAS pages 247 - 248 Ex 1 (f) 1 - 4 & (g) 1 - 3 and
MAS exercise Page 259 Ex 3 (a) –(g)

1. All basic Grade 8 and 9 Geometry (pages 1-7)
2. Congruency
3. Quadrilaterals

Notes on Material for Revision:

Revision exercise A-F and memos are uploaded in GC For Revision F only do up to no.4

Algebra Revision A

1. Calculate:

a) $9+(-14)-(-2)^2$

b) $\sqrt[3]{-64}+2^3$

c) $(-1)^{10}+5-3$

d) $-10-(7)(3)$

e) $5\times\sqrt{\frac{49}{25}}$

f) $\sqrt[3]{\frac{125}{64}}+\frac{1}{6}$

g) $\sqrt{169-144}$

h) $\sqrt{144\times100}$

2. Calculate:

a) $\frac{-(-6)^2-4(-9)}{24}$

b) $\frac{4(3)}{4(3)-(2)(6)}$

c) $4-4\div2$

d) $2^{\frac{1}{2}}$

3. Write as an algebraic expression:

a) Subtract x from the product of 2 and y

b) Square the sum of m and n and divide the result by triple b.

4. Simplify:

a) $-2x^4-2x^4-2x^4$

b) $-x^2-x+1+3x^2-x-2$

Algebra Revision A

MEMO

1. a) $9-14-(4)=-9$

b) $-4+8=4$

c) $1+5-3=3$

d) $-10-21=-31$

e) $5 \times \frac{7}{5}=7$

f) $\frac{5}{4}+\frac{1}{6}=\frac{15+2}{12}=\frac{17}{12}$

g) $\sqrt{25}=5$

h) $12 \times 10=120$

2a) $\frac{-36+36}{24}=\frac{0}{24}=0$

b) $\frac{12}{12-12}=\frac{12}{0}=$ *undefined*

c) $4-\frac{4}{2}=4-2=2$

d) $\frac{1}{2} \div 2 = \frac{1}{2} \times \frac{1}{2}=\frac{1}{4}$

3a) $2y-x$

b) $\frac{(m+n)^2}{3b}$

4a) $-6x^4$

b) $2x^2-2x-1$

Algebra Revision B

Complete the table using your knowledge of the language used in Algebra:

Expression	No. of terms	Type of algebraic expression	Numerical coefficient of 2nd term	Exponent of x in the 1st term	Constant term	Rewrite with descending powers of x
1. $4x - 7x^3 - x^2 + 4$						
2. $-5x^3 + 3x$						
3. $-x - 3x^2 - 7$						
4. $x^5 - x^4 - 3 + 4x$						
5. $-6x^2 - 3x + 1$						
6. $-3x + 7x^4$						
7. $x^3 - x - 5 + 6x^2$						
8. $-x^3$						
9. $3 - 2x$						
10. $7x - 4x^3 - x^2$						

2. Write algebraic expressions for each of the following statements:

(a) the sum of x and y

(b) 8 more than p

(c) the difference between a and b

(d) x more than 6

(e) 12 less than the product of x and y

(f) the square of p

(g) double the value of p

(h) the square root of x

(i) a quarter of a certain number decreased by 3

(j) the cube root of y cubed.

3. How many terms are there in each of the following expressions:

(a) $m + n$

(b) $xy + 7$

(c) $\frac{a+b}{c+d} + 1$

(d) $(a + b)(c + d)$

(e) $3c + 4d \div 2a + 3b$

MEMO Algebra Revision B

Expression	No. of terms	Type of algebraic expression	Numerical Coefficient of 2nd term	Exponent of x in the 1st term	Constant term	Rewrite with descending powers of x
1. $4x - 7x^3 - x^2 + 4$	4	Polynomial	-7	1	4	$-7x^3 - x^2 + 4x + 4$
2. $-5x^3 + 3x$	2	Binomial	3	3	0	$-5x^3 + 3x$
3. $-x - 3x^2 - 7$	3	Trinomial	-3	1	-7	$-3x^2 - x - 7$
4. $x^5 - x^4 - 3 + 4x$	4	Polynomial	-1	5	-3	$x^5 - x^4 + 4x - 3$
5. $-6x^2 - 3x + 1$	3	Trinomial	-3	2	1	$-6x^2 - 3x + 1$
6. $-3x + 7x^4$	2	Binomial	7	1	0	$7x^4 - 3x$
7. $x^3 - x - 5 + 6x^2$	4	Polynomial	-1	3	-5	$x^3 + 6x^2 - x - 5$
8. $-x^3$	1	Monomial	0	3	0	$-x^3$
9. $3 - 2x$	2	Binomial	-2	0	3	$-2x + 3$
10. $7x - 4x^3 - x^2$	3	Trinomial	-4	1	0	$-4x^3 - x^2 + 7x$

2. (a) $x + y$ (b) $p + 8$ (c) $a - b$ (d) $6 + x$ (e) $xy - 12$
   (f) p^2 (g) $2p$ (h) $\sqrt{x}$ (i) $\frac{x}{4} - 3$ (j) $\sqrt[3]{y^3}$

3. (a) 2 (b) 2 (c) 2 (d) 1 (e) 3

INTEGERS REVISION

Calculate:

$-2-10$ 1. $=$	$-2-(-10)$ $=$ 2. $=$	$2-10$ 3. $=$
$-5+2$ 4. $=$	$-5-2$ 5. $=$	$-5-(-2)$ $=$ 6. $=$
$9-7$ 7. $=$	$9-(-7)$ $=$ 8. $=$	$-9-7$ 9. $=$
$-4+6$ 10. $=$	$-4-6$ 11. $=$	$-4-(-6)$ $=$ 12. $=$
$-8+3$ 13. $=$	$-8-3$ 14. $=$	$-8-(-3)$ $=$ 15. $=$
$-8-5$ 16. $=$	$-8+5$ 17. $=$	$1-(-2)$ $=$ 18. $=$
$5-7$ 19. $=$	$-5-(-7)$ $=$ 20. $=$	

MEMO INTEGERS REVISION

1. $-2-10$ $=-12$	2. $-2-(-10)$ $=-2+10$ $=8$	3. $2-10$ $=-8$
4. $-5+2$ $=-3$	5. $-5-2$ $=-7$	6. $-5-(-2)$ $=-5+2$ $=-3$
7. $9-7$ $=2$	8. $9-(-7)$ $=9+7$ $=16$	9. $-9-7$ $=-16$
10. $-4+6$ $=2$	11. $-4-6$ $=-10$	12. $-4-(-6)$ $=-4+6$ $=2$
13. $-8+3$ $=-5$	14. $-8-3$ $=-11$	15. $-8-(-3)$ $=-8+3$ $=-5$
16. $-8-5$ $=-13$	17. $-8+5$ $=-3$	18. $1-(-2)$ $=1+2$ $=3$
19. $5-7$ $=-2$	20. $-5-(-7)$ $=-5+7$ $=2$	

EQUATIONS REVISION

1. Solve the following equations

1. $x+5=9$
2. $y+7=4$
3. $p+4=10$
4. $n+2=-5$
5. $k+12=5$
6. $x+8=-2$
7. $x-4=-2$
8. $y-14=-7$
9. $13=x-15$
10. $x-10=-2$
11. $x-5=-6$
12. $x-11=15$
13. $2=x+9$
14. $1=x-8$
15. $1=x+30$

2. Solve the following equations

1. $2a=-8$
2. $3b=15$
3. $-4c=12$
4. $-5d=-20$
5. $5e=175$
6. $-4h=3$
7. $8k=-2$
9. $3g=-2$
10. $-8i=-1$
11. $\frac{y}{5}=10$
12. $\frac{y}{2}=\frac{3}{4}$
13. $\frac{y}{4}=\frac{1}{2}$
14. $-7j=14$
15. $\frac{y}{5}=5$

3. Solve the following equations

1. $x+4=6$
2. $-x-9=-2$
3. $4-x=7$
4. $5-x=-1$
5. $12-y=-2$
6. $3-y=5$
7. $2x+4=10$
8. $2x-4=-20$
9. $3x+2=-10$
10. $4x-4=16$
11. $5+5x=20$
12. $6x+6=30$
13. $0=12x+60$
14. $17x+34=0$
15. $0=30+5x$
16. $2x-1=0$
17. $-3x+1=4$
18. $5-3x=10$
19. $3x+2=17$
20. $5x-7=28$
21. $28=3p+4$
22. $5x+7=4$
23. $4x+15=9$
24. $3p-4=-20$
25. $28=4-3p$
26. $6x+9=-48$
27. $-x-4=-15$
28. $\frac{4}{x}+4=6$

4. Solve the following equations:

1. $2(3-x)=8$
2. $3(x+2)=2(x+1)$
3. $5(x-1)=4(x-2)$
4. $2(2-x)=3(x+2)$
5. $5(p-1)-(1-2p)=8$
6. $8-3(p+7)=p+5$
7. $1-(x+2)=5+x$
8. $2y-3(y-1)=15-4y$
9. $2(2x+5)=-2(1+x)$
10. $3(x-4)=2(x-2)+2-x$
11. $2-5(1-y)=3y+4$
12. $6(p-1)-p=-2(p+1)+p-1$

EQUATIONS REVISION (MEMO)

1.

1. $x=4$
2. $y=3$
3. $p=6$
4. $n=-7$
5. $k=-7$
6. $x=-10$
7. $x=2$
8. $y=7$
9. $x=28$
10. $x=8$
11. $x=-1$
12. $x=26$
13. $x=-7$
14. $x=9$
15. $x=-29$

2.

1. $a=4$
2. $b=5$
3. $c=-3$
4. $d=4$
5. $e=35$
6. $h=-\frac{3}{4}$
7. $k=-\frac{1}{4}$
8. $f=\frac{1}{2}$
9. $g=-\frac{2}{3}$
10. $i=\frac{1}{8}$
11. $y=50$
12. $y=\frac{3}{2}$
13. $y=2$
14. $j=-2$
15. $y=25$

3.

1. $x=2$
2. $x=7$
3. $x=-3$
4. $x=6$
5. $y=14$
6. $y=-2$
7. $x=3$
8. $x=-8$
9. $x=-4$
10. $x=5$
11. $x=3$
12. $x=4$
13. $x=-5$
14. $x=-2$
15. $x=-6$
16. $x=\frac{1}{2}$
17. $x=-1$
18. $x=-\frac{5}{3}$
19. $x=5$
20. $x=7$
21. $p=8$
22. $x=-\frac{3}{5}$
23. $k=-\frac{3}{2}$
24. $p=-\frac{16}{3}$
25. $p=-8$
26. $x=\frac{-57}{6}$
27. $x=11$
28. $x=2$

4.

1. $x=-1$
2. $x=-4$
3. $x=-3$
4. $x=\frac{-2}{5}$
5. $p=2$
6. $p=-\frac{9}{2}$
7. $x=-3$
1. $y=4$
9. $x=-2$
10. $x=5$
11. $y=\frac{7}{2}$
12. $x=\frac{1}{2}$